

Network Flow Problems

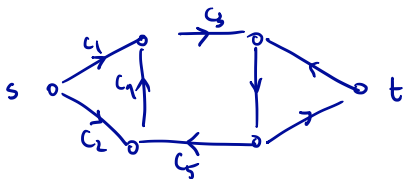
$$G = (V, E)$$

↑
graph
directed graph

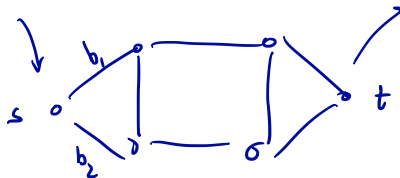
$V = n$ nodes

$E \subseteq V \times V$ with directedness.

shortest path



Max Flow



Flow Conservation Constraints

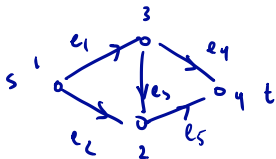
A = node-arc incidence matrix

A is $|V| \times |E|$ matrix.

Encoding : $G = (V, E)$

$$A_{ie} = \begin{cases} +1 & \text{if arc } e \text{ leaves node } i \\ -1 & \text{" " enters " } \\ 0 & \text{o.w.} \end{cases}$$

Node-Arc incidence matrix: example



e_1	e_2	e_3	e_4	e_5
1	1	0	0	0
0	-1	-1	0	1
-1	0	1	1	0
0	0	0	-1	-1

Flow conservation at node i :

$$f \in \mathbb{R}^E :$$

$$\boxed{a_i^T f = 0}$$

Shortest Path

Flow of unit 1 and minimal
cost from s to t :

Flow Cons: $Af = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} s \\ t \end{matrix}$ Check: redundant
so can omit one.

$$\min: c^T f$$

$$\text{st: } Af = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{row } t \text{ omitted}$$

$$f \geq 0$$

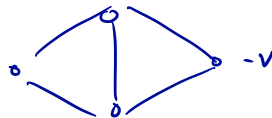
"only one path selected"

Shortest Path
LP

Max Flow

Instead of cuts, we have capacities.

$$\begin{aligned} \max: & v \\ \text{st: } & Af + d v = 0, \quad d = \begin{cases} -1 & i=s \\ +1 & i=t \\ 0 & \text{otherwise} \end{cases} \\ & f \leq b \leftarrow \text{capacity vector}, \\ & f \geq 0 \end{aligned}$$



Max Flow LP